Chapter 33 Review

Campbell Biology, AP Edition, 12e

Chapter 33 Review

Summary of Key Concepts

The vast majority of animals are invertebrates, which have niches in every ecosystem on Earth. Justify the claim that one group of invertebrates, dung beetles, has a huge ecological impact. **(BIG IDEAS 2 & 4)**

This table recaps the animal groups surveyed in this chapter.

Key Concept	Phylum	Description
CONCEPT 33.1 **Sponges are basal animals that lack tissues** ? *Lacking tissues and organs, how do sponges accomplish tasks such as gas exchange, nutrient transport, and waste disposal?*	Porifera (sponges)	Lack tissues; have choanocytes (collar cells—flagellated cells that ingest bacteria and tiny food particles)
CONCEPT 33.2 **Cnidarians are an ancient phylum of eumetazoans** ? *Describe the cnidarian body plan and its two major variations.*	Cnidaria (hydras, jellies, sea anemones, corals)	Unique stinging structures (nematocysts) housed in specialized cells (cnidocytes); diploblastic; radially symmetrical; gastrovascular cavity (digestive compartment with a single opening)
CONCEPT 33.3 **Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms** ? *Is the lophotrochozoan clade united by unique morphological features shared by all of its members? Explain.*	Platyhelminthes (flatworms)	No body cavity; dorsoventrally flattened; gastrovascular cavity or no digestive tract
	Syndermata (rotifers and acanthocephalans)	Hemocoel; rotifers have alimentary canal (digestive tube with mouth and anus) and jaws (trophi); acanthocephalans are parasites of vertebrates
	Ectoprocta and Brachiopoda	Coelom; have lophophores (feeding structures bearing ciliated tentacles)
	Mollusca (clams, snails, squids)	Hemocoel; reduced coelom; three main body parts (muscular foot, visceral mass, mantle); most have hard shell made of calcium carbonate
	Annelida (segmented worms)	Coelom; body wall and internal organs are segmented (except digestive tract, which is unsegmented)
CONCEPT 33.4 **Ecdysozoans are the most species-rich animal group** ? *Describe some ecological roles of nematodes and arthropods.*	Nematoda (roundworms)	Hemocoel; cylindrical body with tapered ends; no circulatory system; undergo ecdysis
	Arthropoda (spiders, centipedes, crustaceans, and insects)	Hemocoel; reduced coelom. Have segmented body, jointed appendages, and exoskeleton made of protein and chitin
CONCEPT 33.5 **Echinoderms and chordates are deuterostomes** ? *You've read that echinoderms and chordates are closely related and have evolved independently for over 500 million years. Explain how both of these statements can be correct.*	Echinodermata (sea stars, sea urchins)	Coelom; bilaterally symmetrical larvae and five-part body organization as adults; unique water vascular system; endoskeleton
	Chordata (lancelets, tunicates, vertebrates)	Coelom; have notochord; dorsal, hollow nerve cord; pharyngeal slits; post-anal tail (see Figure 34.3)

Metazoa; Eumetazoa; Bilateria; Lophotrochozoa; Ecdysozoa; Deuterostomia

An evolutionary tree is given along the table. All 11 rows are metazoa. The bottom 10 are eumetazoa. The bottom 9 are bilateria. Rows 3 through 7 are lophotrochozoa. Rows 8 and 9 are ecdysozoa. Rows 10 and 11 are deuterostomia.

Key concept	Phylum	Description
Concept 33.1. Sponges are basal animals that lack tissues (pages 690 to 691). Lacking tissues and organs, how do sponges accomplish tasks such as gas exchange, nutrient transport, and waste disposal?	Porifera (sponges)	Lack tissues; have choanocytes (collar cells - flagellated cells that ingest bacteria and tiny food particles)
Concept 33.2. Cnidarians are an ancient phylum of eumetazoans (pages 691 to 693). Describe the cnidarian body plan and its two major variations.	Cnidaria (hydras, jellies, sea anemones, corals)	Unique stinging structures (nematocysts) houses in specialized cells (cnidocytes); diploblastic; radially symmetrical; gastrovascular cavity (digestive compartment with a single opening)
Concept 33.3. Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms (pages 694 to 705). Is the lophotrochozoan clade united by unique morphological features shared by all of its members? Explain.	Platyhelminthes (flatworms)	No body cavity; dorsoventrally flattened; gastrovascular cavity or no digestive tract
Concept 33.3.	Syndermata (rotifers and acanthocephalans)	Hemocoel; rotifers have alimentary canal (digestive tube with mouth and anus) and jaws (trophi); acanthocephalans are parasites of vertebrates
Concept 33.3.	Lophophorates: ectoprocta, brachiopoda	Coelom; have lophophores (feeding structures bearing ciliated tentacles)
Concept 33.3.	Mollusca (clams, snails, and squids)	Hemocoel; reduced coelom; three main body parts (muscular foot, visceral mass, mantle); most have a hard shell made of calcium carbonate
Concept 33.3.	Annelida (segmented worms)	Coelom; body wall and internal organs are segmented (except digestive tract, which is unsegmented)

Key concept	Phylum	Description
Concept 33.4. Ecdysozoans are the most species-rich animal group (pages 705 to 713). Describe some ecological roles of nematodes and arthropods.	Nematoda (roundworms)	Hemocoel; cylindrical body with tapered ends; no circulatory system; undergo ecdysis
Concept 33.4.	Arthropoda (spiders, centipedes, crustaceans, and insects)	Hemocoel; reduced coelom. Have segmented body, jointed appendages, and exoskeleton made of protein and chitin
Concept 33.5. Echinoderms and chordates are deuterostomes (pages 713 to 715). You've read that echinoderms and chordates are closely related and have evolved independently for over 500 million years. Explain how both of these statements can be correct.	Echinodermata (sea stars, sea urchins)	Coelom; bilaterally symmetrical larvae and five-part body organization as adults; unique water vascular system; endoskeleton
Concept 33.5.	Chordata (lancelets, tunicates, vertebrates)	Coelom; have notochord; dorsal hollow nerve cord; pharyngeal slits; post-anal tail (see Figure 34.3)

*For suggested answer, see*Appendix A.**